SOLAR AGRICULTURAL AUTHORITY · EVIDENCE SPECIALIST MISSION

CASE 01 · ISS GREENHOUSE MODULE

Campaign 1 · Low Earth Orbit · Gravitropism in Microgravity

Agent:	**Date:**	**Class/Period:**

Mission question: Why are the lettuce roots tangled and directionless even though water, nutrients, temperature, and lighting appear normal?

SCIENCE FOCUS: GRAVITROPISM IN MICROGRAVITY

Plants normally use gravity to help guide root and shoot growth. In microgravity, that directional cue becomes unreliable, so engineers must provide another way for plants to orient.

Science status: The ISS case story is fictional. Gravitropism, statoliths, and plant responses to environmental cues are real.

1 · Vocabulary

Gravitropism	Directional growth in response to gravity.	**Microgravity**	Apparent weightlessness during continuous free fall in orbit.
Statocyte	Plant cell involved in sensing gravity.	**Statolith**	Dense starch-filled structure that helps provide a direction cue.
Phototropism	Directional growth in response to light.		

2 · Initial thinking

What evidence would help you distinguish a resource problem from an orientation problem?

3 · Investigate four evidence sources

Record only evidence that helps support or reject a cause. O = observation/data; I = inference/explanation.

Source	Observation or data	What does it support or reject?	O / I
Crew			
Sensors			
Plants			
Mission logs			

4 · Test the competing explanations

Explanation	Status	One evidence-based reason
Nutrient solution is damaging roots.		
The light cycle is wrong.		
The seed stock is defective.		
Microgravity disrupts orientation.		

DIAGNOSIS & DESIGN RESPONSE

INDIVIDUAL PRODUCT

Use evidence from the game. You may reopen sources while writing.

5 · Build the mechanism

Earth gravity	Microgravity
Statoliths ________________ → consistent direction signal → roots grow ________________	Statoliths do not ________________ consistently → unreliable direction signal → roots ________________

6 · Diagnosis

State the cause in your own words. Then reject the most tempting alternative.

Diagnosis	Rejected alternative + evidence

7 · Claim-Evidence-Reasoning

CLAIM	
EVIDENCE	
REASONING	

8 · Engineering response

Choose or design a consistent orientation cue: root channels/mesh, moisture gradient, shoot lighting plus a root guide, rotating chamber, or another justified design.

Design	One criterion	One constraint

9 · Exit ticket

Engineers add narrow root channels but do not change the lighting. Which symptom should improve first: tangled roots or pale leaves? Explain.

Optional extension: Compare a physical root guide with a rotating growth chamber. Which is easier to use on the ISS, and which more closely replaces gravity?

Student source line: NASA Plant Biology and Plant Gravity Perception resources; NGSS MS-ETS1 Engineering Design.